INFOMDW: Data Wrangling

Data Normalization and Discretization

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Reading Material for Today

- Data Mining: Concepts and Techniques Book

  CH 3.5



Topics for Today

- Data transformation
- Data Integration
- Data reduction
- Data discretization

Data Transformation

Data Transformation

- A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values
- Methods for data transformation
  - Smoothing: Remove noise from data
  - Attribute/feature construction
    - New attributes constructed from the given ones
  - Aggregation: Summarization, data cube construction

Data Transformation (Cont.)

- Methods for data transformation
  - Normalization: Scaled to fall within a smaller, specified range
    - Min-max normalization
    - Z-score normalization
    - Normalization by decimal scaling
  - Data reformatting:
    - E.g. Jack Wilsher ➔ Wilsher, J.
  - Use the same unit:
    - Records in inches and cm
    - Records with prices in Euros and Dollars

Data Normalization (Standardization)

The goal of standardization or normalization is to make an entire set of values have a particular property.

Data Normalization – Min-Max Normalization

- Transform the data from a given range with $[min_A, max_A]$ to a new interval $[new_min_A, new_max_A]$ for a given attribute A :

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

where v is the current value of attribute A.

Data Normalization – Min-Max Normalization

- Example:

  Suppose that the minimum and the maximum in the attribute income are €12,000 and €98,000, respectively

  We would like to map the income into the interval $[0,1]$

  Using min-max normalization, a value of €73,600 for income is transformed into:

$$\frac{73{,}600 - 12{,}000}{98{,}000 - 12{,}000}(1.0 - 0.0) + 0.0 = 0.716$$

Data Normalization – Z-Score Normalization

- Transform the data by converting the values to a common scale with an average of zero and a standard deviation of one.
- A value, v, of attribute A is normalized to v' by computing:

$$v' = \frac{v - \bar{A}}{\sigma_A}$$

where $\bar{A}$ and σ_A are the mean and standard deviation of attribute A, respectively.

Data Normalization – Z-Score Normalization

- Example:
  - Suppose that the mean and standard deviation of the values for the feature income are 54,000 and 16,000, respectively. With z-score normalization, a value of €73,600 for income is transformed to:

$$\frac{73{,}600 - 54{,}000}{16{,}000} = 1.225$$

Data Normalization – Decimal Scaling Normalization

- Transform the data by moving the decimal points of values of attribute A.
- The number of decimal points moved depends on the maximum absolute value of A.
- A value v of A is normalized to v' by computing: $v' = \frac{v}{10^j}$

  where j is the smallest integer such that $\max(|v'|) < 1$

Data Normalization – Decimal Scaling Normalization

- Example:
  - Suppose that the recorded values of A range from −986 to 917.
  - The maximum absolute value of A is 986.
  - To normalize by decimal scaling, we divide each value by 1,000 (i.e., $j = 3$) so that −986 normalizes to −0.986 and 917 normalizes to 0.917

Data Transformation (Cont.)

- Exercise
  - Use flash fill of Microsoft Excel to convert set of names from the format Family_name, First_name middle_initial. to First_name Family_name
    - E.g. Wilsher, John K. to John Wilsher

Entity Resolution

Data Integration

- Combines data from multiple sources into a coherent store
- Schema integration: e.g., A.cust-id ≡ B.cust-No
  - Integrate metadata from different sources
- Entity identification problem:
  - Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton
- Detecting and resolving data value conflicts
  - For the same real-world entity, attribute values from different sources are different
  - Possible reasons: different representations, different scales, e.g., metric vs. British units

Handling Redundancy when Integrating Data

- Handling data redundancy is an important task of the data integration
- Duplicate records can be identified by applying entity resolution techniques
- Redundant attributes can be detected by correlation and covariance analysis
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining quality

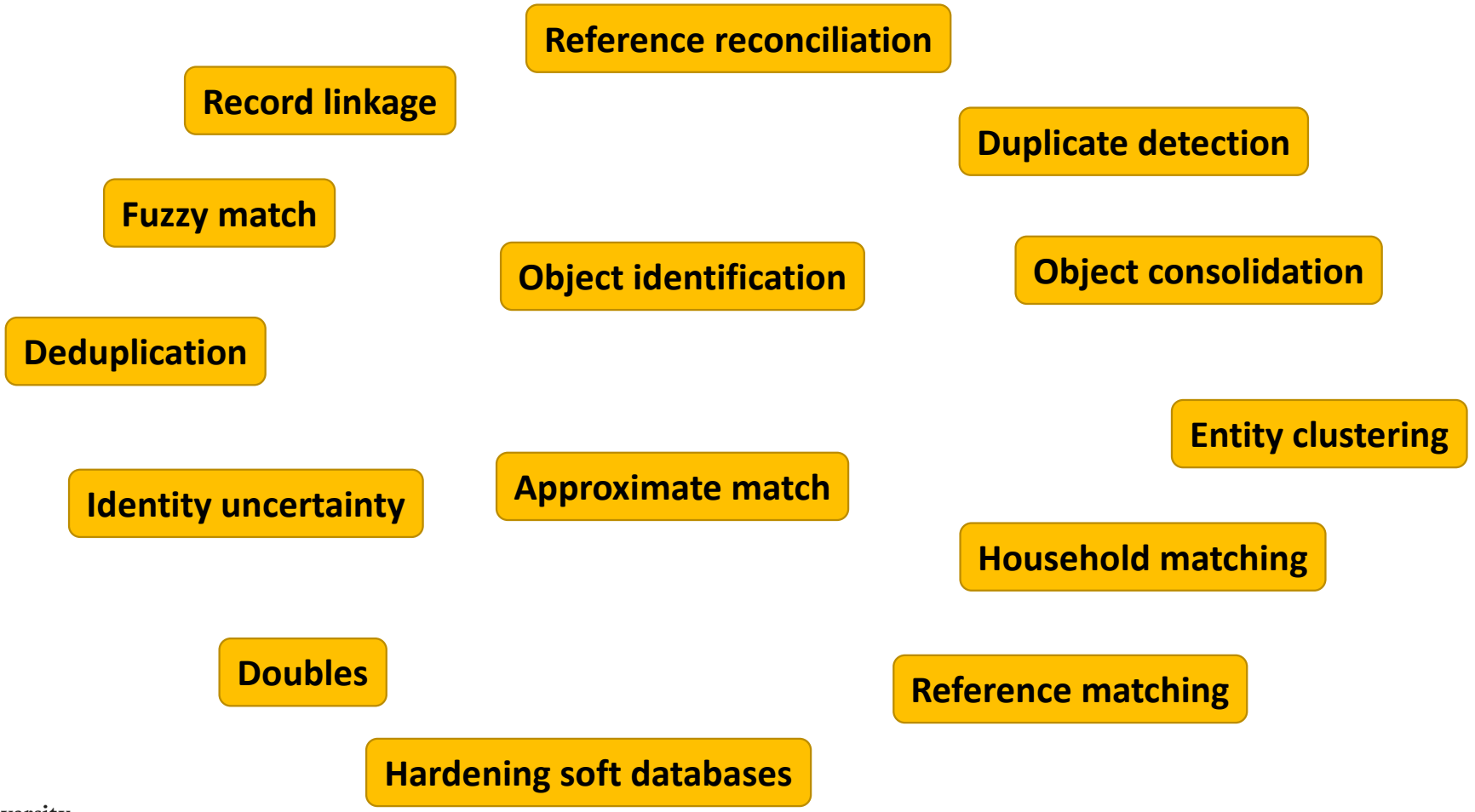
Entity Resolution

Problem of identifying and linking/grouping different representations of the same real-world object.

Examples:

- Different ways of addressing (names, email addresses, FaceBook accounts) the same person in text.
- Web pages with differing descriptions of the same business.
- Different photos of the same object.
- …

Ironically, Entity Resolution has Many Duplicate Names



Entity Resolution – Normalization

- Schema Normalization
  - Schema matching: e.g., Contact No. vs. Phone
  - Compound attributes: e.g., address vs. (street, city, zip)
- Data Normalization
  - Capitalization, white-space normalization
  - Correcting typos, replacing abbreviations, variations, nick names
    - Usually done by employing dictionaries

Entity Resolution – Matching Features

- Given two records, compute a vector of similarity scores for corresponding features
- E.g., to match two bibliographical references, compute:
  - First author match score
  - Title match score
  - Venue match score
  - Year match score
  - …
- Score can be Boolean (match/mismatch) or a continuous value based on specific similarity measure (distance function).

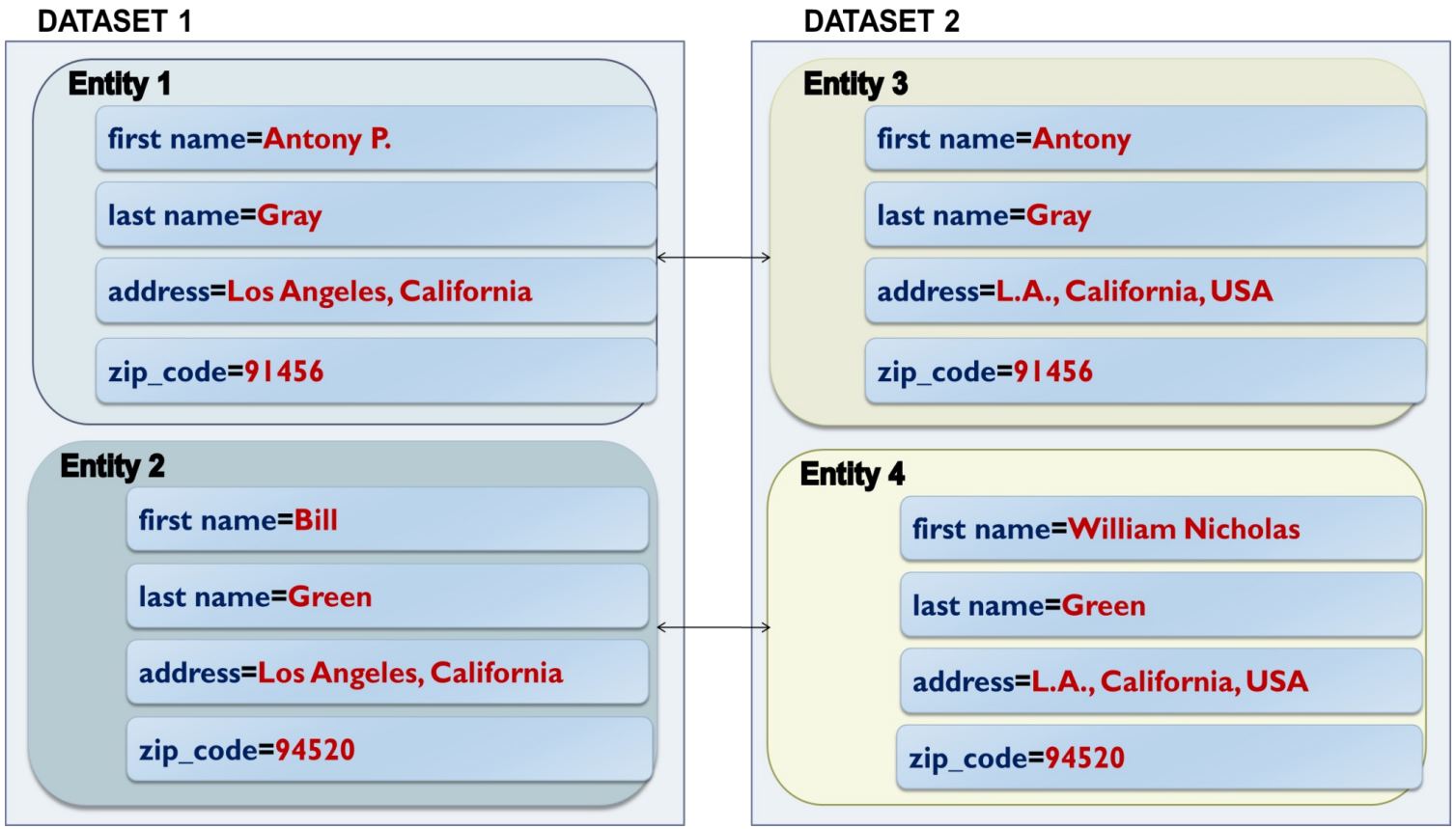
Entity Resolution – Similarity Measures

- Atomic similarity measures
  - Difference between numerical values
  - Jaro for comparing names
  - Edit distance for typos
  - Phonetic-based
    - Soundex
- Set similarity
  - Jaccard similarity
- Vector-based
  - Cosine similarity

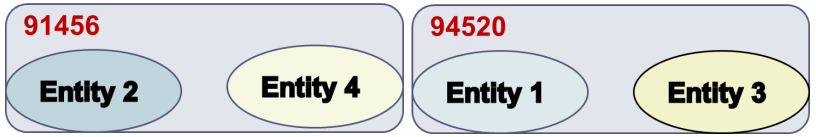
Entity Resolution – Issues

- Similarity measures has different scales
  - How to combine the different values to compare the tuples
- Pairwise similarity between records is expensive
  - Takes $O(n^2)$
- Blocking to reduce the quadratic time complexity
  - Divide the records into blocks
  - Perform pairwise comparison between records within the same block only
  - When $\#blocks = k$ and block size $\approx (n/k)$, the time complexity $O(k(n/k)^2)$

Standard Blocking – Example

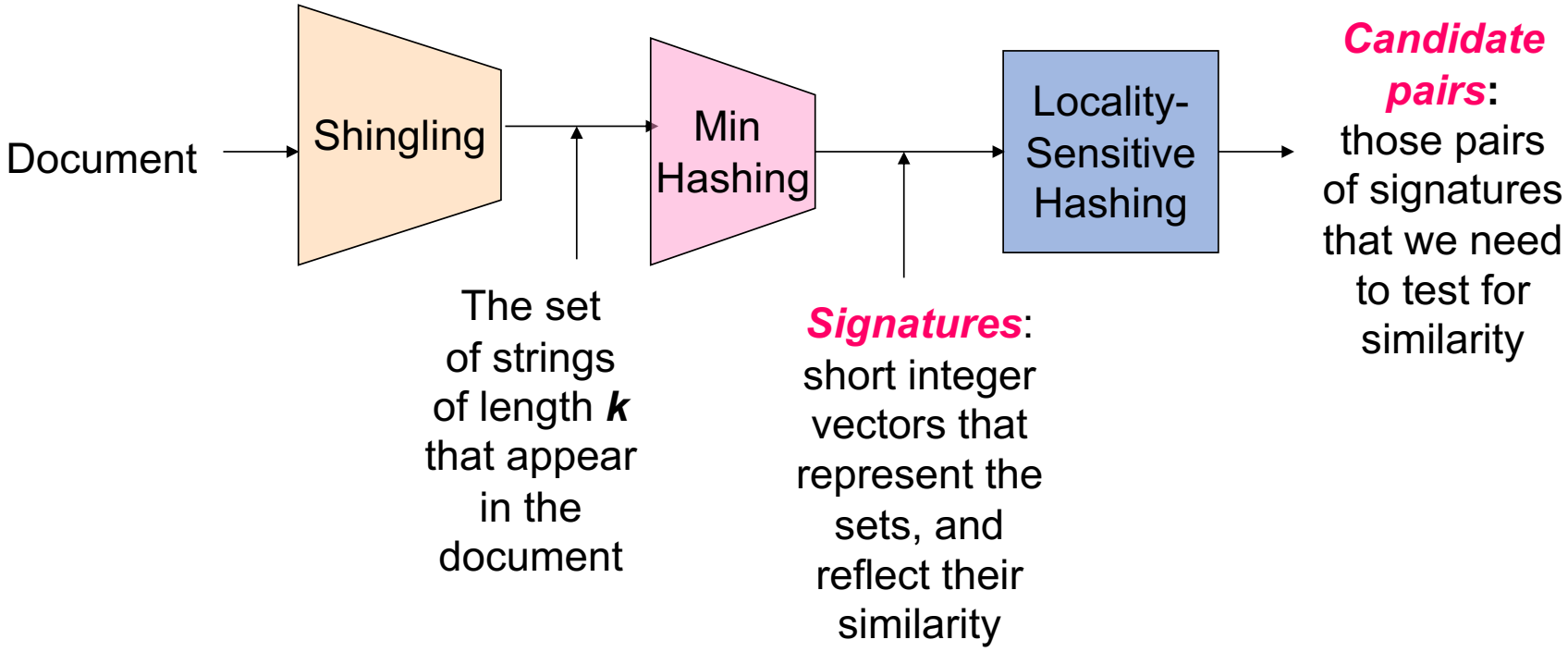


Blocks on zip_code:



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3 Essential Steps for Similar Docs



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Example

- **Rows** = elements (shingles)
- **Columns** = sets (documents)
  - 1 in row ***e*** and column ***s*** if and only if ***e*** is a member of ***s***
  - Column similarity is the Jaccard similarity of the corresponding sets (rows with value *1)*
  - **Typical matrix is sparse!**
- **Data**
  - **D1 = {jumped}**
  - **D2 = {bumped}**
  - **D3 = {dumped}**
  - **D4 = {pumped}**

Documents

Shingles

	D1	D2	D3	D4
bum	0	1	0	0
dum	0	0	1	0
jum	1	0	0	0
mpe	1	1	1	1
ped	1	1	1	1
pum	0	0	0	1
ump	1	1	1	1

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Hashing

- **Key idea:** "hash" each column ***C*** to a small ***signature h(C)***, such that:
  (1) ***h(C)*** is small enough that the signature fits in RAM
  (2) ***sim(C_1, C_2)*** is the same as the "similarity" of signatures ***h(C_1)*** and ***h(C_2)***
- **Goal: Find a hash function *h(·)* such that:**
  - If ***sim(C_1,C_2)*** is high, then with high prob. ***h(C_1) = h(C_2)***
  - If ***sim(C_1,C_2)*** is low, then with high prob. ***h(C_1) ≠ h(C_2)***
- **Hash docs into buckets. Expect that "most" pairs of near duplicate docs hash into the same bucket!**

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Min-Hashing

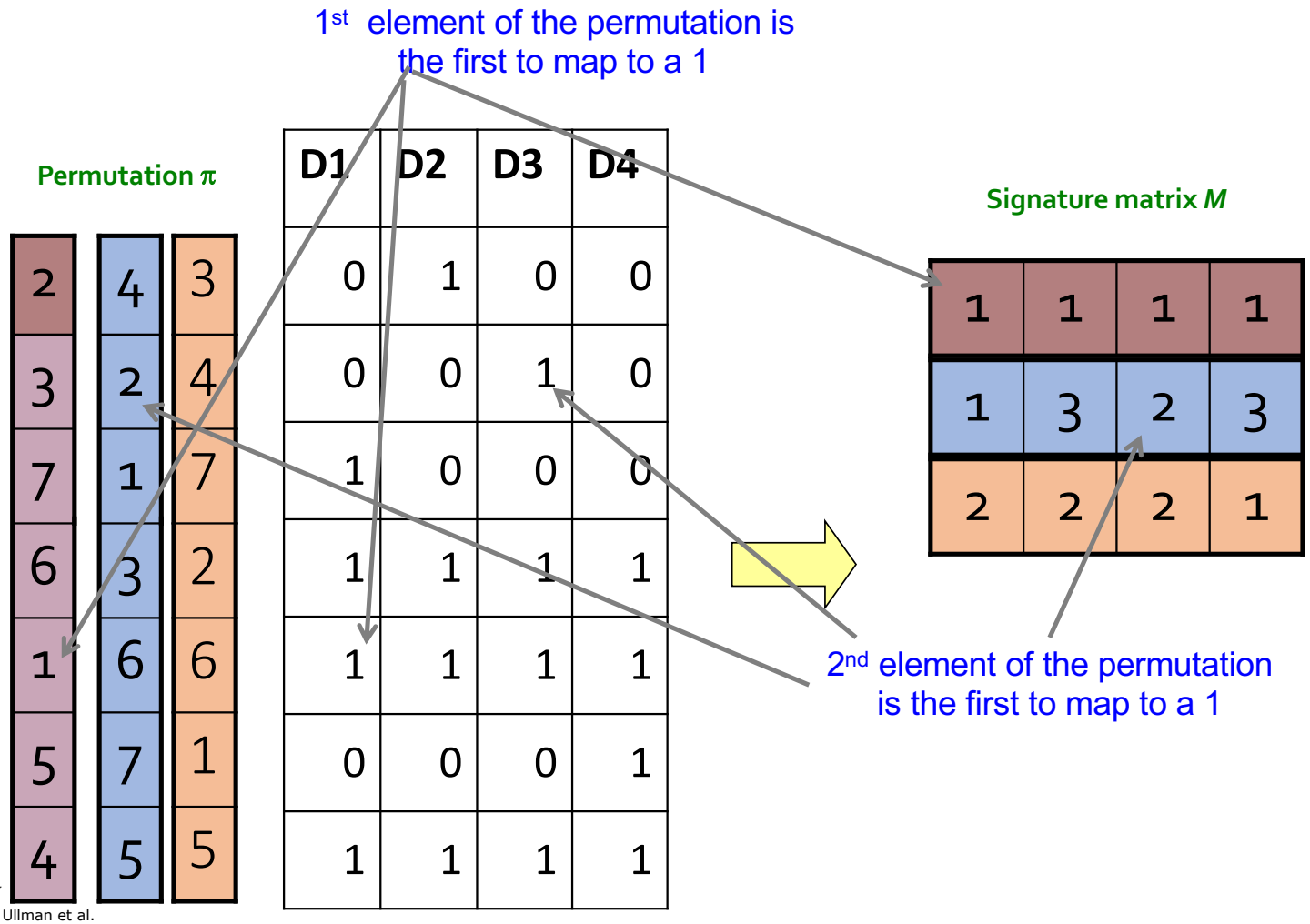
- Imagine the rows of the Boolean matrix permuted under **random permutation π**
- Define a **"hash" function $h_\pi(C)$** = the index of the **first** (in the permuted order π) row in which column ***C*** has value **1**:

$$h_\pi(C) = min_\pi \pi(C)$$

- Use several (e.g., 100) independent hash functions (i.e., permutations) to create a signature of a column

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Min-Hashing Example



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Similarity of Signatures

- Clearly: $\Pr[h_\pi(C_1) = h_\pi(C_2)] = sim(C_1, C_2)$
- Now generalize to multiple hash functions

- **The *similarity of two signatures* is the fraction of the hash functions in which they agree**

- **Note:** Because of the Min-Hash property, the similarity of columns is the same as the expected similarity of their signatures

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LSH for Min-Hash

- **Big idea: Hash columns of signature matrix *M* several times**
- Arrange that (only) **similar columns** are likely to **hash to the same bucket**, with high probability
- **Candidate pairs are those that hash to the same bucket**

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Standard Blocking

Algorithm:

1. Select the most appropriate attribute name(s) w.r.t. noise and distinctiveness.
2. Transform the corresponding value(s) into a Blocking Key (BK)
3. For each BK, create one block that contains all entities having this BK in their transformation.

[Fellegi et. al., JASS 1969]

Works as a hash function! → Blocks on the **equality** of BKs

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Data Discretization

Data Discretization

- Three types of attributes
  - Nominal: values from an unordered set, e.g., color, profession
  - Ordinal: values from an ordered set, e.g., military or academic rank
  - Numeric: real numbers, e.g., integer or real numbers
- Discretization: Divide the range of a continuous attribute into intervals
  - Interval labels can be used to replace actual data values
  - Supervised vs. unsupervised
  - Split (top-down) vs. merge (bottom-up)

Discretization Methods

- Typical methods: All the methods can be applied recursively
  - Binning – Histograms
  - Clustering
  - Classification (e.g. Decision-trees)
  - Correlation

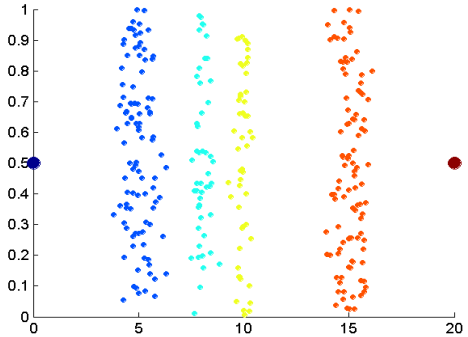
Discretization Methods – Binning

- Equal-width (distance) partitioning
  - Divides the range into N intervals of equal size: uniform grid
  - If *A* and *B* are the smallest and largest values of the attribute, the width of intervals will be: $W = (B - A)/N$.
  - The most straightforward, but outliers may dominate presentation
  - Skewed data is not handled well

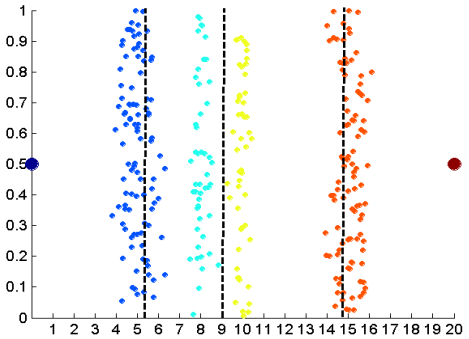
Discretization Methods – Binning (Cont.)

- Equal-depth (frequency) partitioning
  - Divides the range into *N* intervals, each containing approximately same number of samples
  - Good data scaling
  - Managing categorical attributes can be tricky

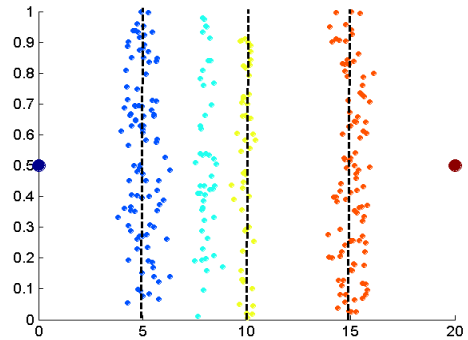
Discretization Methods – Binning (Cont.)



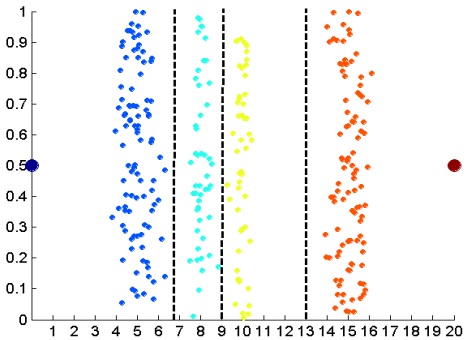
Data



Equal frequency (binning)



Equal width (binning)

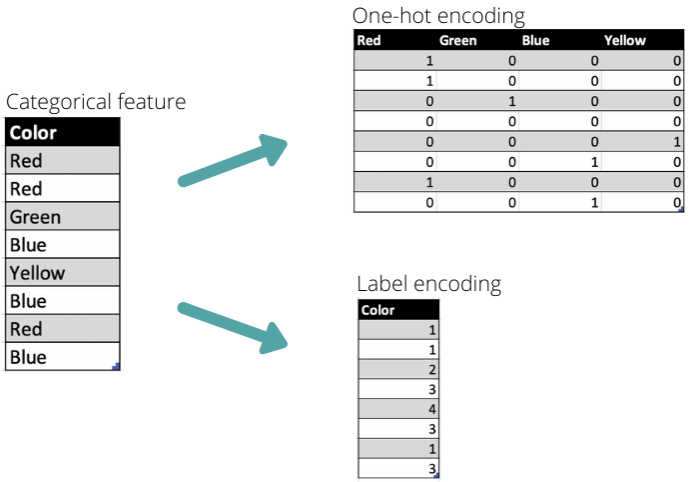


Clustering-based discretization leads to better results (for this dataset)

Discretization Methods – Classification & Correlation

- Classification (e.g., decision tree analysis)
  - Supervised: Given class labels, e.g., cancerous vs. benign
  - Using entropy to determine split point (discretization point)
  - Top-down, recursive split
- Correlation analysis
  - Supervised: use class information
  - Bottom-up merge: find the best neighboring intervals (those having similar distributions of classes) to merge
  - Merge performed recursively, until a predefined stopping condition is satisfied

Other Data Preparation Techniques



- Transforming categorical data into numerical data (See the Figure)
  - Label encoding
  - One-hot encoding
- Data Enrichment
  - Augmentation
  - Join & Union
- Data Validation
  - Check Permitted Characters
  - Find Type-Mismatched Data

Wrap-Up

Summarize what you learned today in 2-minutes



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